LAB MANUAL

User-Defined Functions and ArrayList (C#)

Course: Programming Fundamentals / Object-Oriented Programming

Topic: User-Defined Functions and ArrayList

Activity Type: Laboratory Exercises

Level: Basic

I. Laboratory Objectives

At the end of this laboratory session, the students should be able to:

1. Create simple user-defined functions in C#.
2. Declare and use an ArrayList.
3. Pass an ArrayList as a parameter to a function.
4. Display ArrayList contents using a user-defined function.

II. Laboratory Requirements

- Computer with C# compiler / IDE (Visual Studio or similar)
- Basic knowledge of C# syntax
- Namespace: `System.Collections`

III. Laboratory Rules

- Use `void` user-defined functions only.
- No user input is required.
- No return values are allowed.
- Focus on correct syntax and proper function calls.

IV. Laboratory Exercises

🧪 LAB SET 1: INTRODUCTORY ACTIVITIES

Exercise 1: Display Colors

Objective: To display ArrayList elements using a function.

Instructions: 1. Create an `ArrayList` that stores three (3) colors. 2. Create a user-defined function named `ShowColors`. 3. Pass the ArrayList to the function. 4. Display all colors inside the function.

Exercise 2: Display Numbers

Objective: To pass numeric data stored in an ArrayList to a function.

Instructions: 1. Create an `ArrayList` that stores four (4) numbers. 2. Create a function named `ShowNumbers`. 3. Display all numbers using a loop inside the function.

Exercise 3: Display Fruits

Objective: To display string elements using a user-defined function.

Instructions: 1. Create an `ArrayList` that stores three (3) fruits. 2. Create a function named `ShowFruits`. 3. Call the function once from `Main()`.

🧪 LAB SET 2: PRACTICE ACTIVITIES

Exercise 4: Display Names

Objective: To reinforce passing an ArrayList to a function.

Instructions: 1. Create an `ArrayList` that stores five (5) names. 2. Create a function named `DisplayNames`. 3. Use a `foreach` loop to display all names.

Exercise 5: Display Subjects

Objective: To display multiple values stored in an ArrayList.

Instructions: 1. Create an `ArrayList` that stores four (4) school subjects. 2. Create a function named `ShowSubjects`. 3. Display all subjects inside the function.

Exercise 6: Display Hobbies

Objective: To practice function calls with ArrayList parameters.

Instructions: 1. Create an `ArrayList` that stores three (3) hobbies. 2. Create a function named `ShowHobbies`. 3. Display each hobby on a new line.

🧪 LAB SET 3: REINFORCEMENT ACTIVITIES

Exercise 7: Display Animals

Objective: To strengthen understanding of ArrayList traversal.

Instructions: 1. Create an `ArrayList` that stores four (4) animals. 2. Create a function named `ShowAnimals`. 3. Display all animals using a loop.

Exercise 8: Display Cities

Objective: To consistently apply function and ArrayList concepts.

Instructions: 1. Create an `ArrayList` that stores three (3) cities. 2. Create a function named `ShowCities`. 3. Display all cities inside the function.

Exercise 9: Display Favorite Things

Objective: To apply learned concepts independently.

Instructions: 1. Create an `ArrayList` that stores five (5) favorite things. 2. Create a function named `ShowFavorites`. 3. Display all items using the function.

V. Submission Guidelines

- Submit one (1) C# source file per student.
- File name format: `Lastname_Firstname_Lab.cs`
- Ensure the program runs without errors.

VI. Evaluation Criteria

Criteria	Percentage
Correct use of ArrayList	40%
Correct function implementation	40%
Program output correctness	20%

VII. Conclusion

This laboratory activity introduces the basic use of user-defined functions together with ArrayList in C#. Mastery of these concepts is essential before proceeding to advanced data structures and function operations.